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Under pressure: Integrating machine learning to quantify clutch moments in American football

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Abstract

This study introduces a novel pressure index framework designed to quantify the intensity of crucial moments in an American football game. Our model integrates game-specific metrics such as time remaining, score differential, field position, and play context to establish quantitative measures of pressure on offensive and defensive units. The two proposed indices, viz, the composite offensive pressure index (PIO3) and defensive pressure index (PID) dynamically capture the evolving situational stress during a game. To validate and assess their predictive power, we employ machine learning techniques, including Logistic Regression, Random Forest, and Gradient Boosting models. Using play-by-play data from the 2023 NFL season, the results demonstrate that the ensemble learning methods outperform logistic regression, with Gradient Boosting achieving the highest predictive precision (AUC = 0.944) in forecasting drive success. This machine learning-driven framework provides new insights into situational dynamics and performance under pressure such as identifying pivotal clutch moments and quantifying critical drive transitions offering potential applications in advanced performance analysis, coaching strategies, and sports analytics research.

Keywords

Decision-making, game-specific metrics, National Football League, sport analytics

MSC 2020 Subject Classification

91A10, 91B12, 62P10, 68T07

Introduction

In recent years, the field of sports analytics has experienced significant advancements, driven by the proliferation of rich data sets and breakthroughs in statistical and machine learning techniques.[1] American football, in particular, has become a focal point for quantitative analyses aimed at understanding game dynamics, optimizing player performance, and refining strategic decision-making.[2] Despite these developments, the challenge of quantitatively measuring situational pressure, especially during high-stakes, clutch moments, remains largely unexplored, as traditional metrics do not fully capture the complexities of decision-making under pressure.[3]

Pressure in sports is often linked to pivotal moments that can dramatically influence the outcome of a game. In cricket, for example, pressure builds as teams face a decrease in the number of balls and the potential loss of key wickets, leading to an increasing urgency to achieve target scores.[4] Similarly, in American football, each match unfolds as a series of strategic drives where every play is influenced by unique conditions such as the current down, yards to go, field position, and remaining time. These key terms define the framework within which teams operate: the 'down' system dictates the number of opportunities available to maintain or extend a drive, while 'field position' determines the distance that must be overcome to score. This dynamic interplay creates discrete phases of play where the pressure on both offense and

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defense evolves continuously, underscoring the need for detailed metrics that capture these shifting conditions.

This paper introduces a novel pressure index framework designed to quantify situational pressure in American football. The proposed Composite Offensive Pressure Index (PIO3) and Defensive Pressure Index (PID) integrate game-specific variables such as time remaining, score differential, field position, and play context, to develop robust measures of pressure on both offensive and defensive units. To validate these indices, we leverage machine learning techniques including Logistic Regression, Random Forest, and Gradient Boosting models to assess their predictive power in forecasting drive success.

This study contributes to the field in several keyways. First, it introduces a quantitative framework that systematically measures situational pressure in American football, filling a critical gap in sports analytics research. Second, by leveraging machine learning methodologies, we demonstrate how advanced analytics can effectively predict the drive success based on pressure- related metrics, offering a data-driven approach to evaluating high-pressure situations. Finally, the framework has practical applications for coaching strategies, player evaluation, and broader research into the role of pressure across competitive sports environments.

The remainder of this paper is structured as follows. Section "Related work" reviews relevant literature on sports analytics and pressure quantification. Section "Methodology" details the development of the pressure indices, selection of game-specific metrics, and integration of machine learning techniques. Section "Data analysis" presents empirical analysis using historical NFL data, compares model performances, and discusses key findings. Finally, Section "Conclusion and future work" concludes the paper with a summary of contributions and potential directions for future research.

By developing a systematic approach to quantifying situational pressure, this study provides a powerful analytics-driven tool for understanding decision-making and performance under high- stakes conditions in American football. The findings have implications not only for football but also for broader sports analytics research, paving the way for future investigations into pressure dynamics in various competitive settings.

Related work

The quantification of situational pressure in sports has become an increasingly prominent area within sports analytics, with research spanning basketball, baseball, football (soccer), cricket etc. In American football, traditional models such as win probability and expected points added (EPA) have provided valuable insights into game dynamics, strategic decision-making, and player performance. However, these models often focus on aggregate outcomes and may not fully capture the nuanced, moment-to-moment pressure experienced by teams and players during critical plays.

The first attempt to quantify the concept of pressure is the work by Baumeister.[5] Though the work is more in proximity to social psychology than to sports, the results were very generalized. It tells that as the pressure increases the consciousness of the subject's process of performance and accordingly the increasing attention disrupts the automatic or overlearned nature of execution. Strauss[6] in his work connecting choking under pressure found that in team games when the public expects team success but not the person performing then the level of performance of players decreases.

Sarlis et al.[7] applied data mining tools to measure the performance of individuals and teams in the NBA in different clutch situations towards the last few minutes of matches using a novel tool which they called Estimation of Clutch Competency (EoCC) metric. In the field of basketball, Otten[8] applied structural equation modelling (SEM) on personality traits and situational influences of 201 players and established that the performance of the players got devalued as self-focus increased. Downey and McGarrity[9] studied how the performance of baseball players especially the pitchers during different match situations. The study finds that while the pitchers can keep the match proceedings in their control in low-pressure situations, increased cognitive load increases pressure on the pitchers and accordingly hampers their performance. Hibbs[10] is another work where the author dealt with performance under pressure in the field of baseball. The paper finds that the awareness of the clutch situation, the ability to experience related stress, care for the contest's outcome, and skill are the key determinants of decision-making under pressure.

Not surprisingly there are several works on performance under pressure situation or their quantification on the world's most popular game, football (soccer). An interesting and authoritative review paper in this regard is that of Pourreza et al.[11] The paper by Seifried and Pa-patheodorou,[12] though not an analytical work provides guidelines to couches on how to train footballers to perform well in clutch situations by controlling emotions, coping with different situations and practicing under game situations in critical moments. As expected, some of the papers that have addressed the pressure of penalty kicks and the psychological factors that influence the performance of footballers have enriched the literature viz. Arrondel et al.,[13] Gu et al.,[14] Navia et al.[15] etc.

In cricket, the original attempt to quantify choking by teams was by Lemmer,[16] by combining the required run rate and resource used at the end of each over of a cricket match. Modifying the index further Bhattacharjee and Lemmer[4] developed the pressure index (PI3) in cricket to quantify the pressure experienced by the team batting

second in a limited over cricket match. The paper also presents how PI3 can be used to quantify the performance of cricketers factored by the difficulty level at which they are performing. A similar attempt was also made in cricket by Shah and Shah[17].

For a systematic review on quantifying performance of individual players and teams in different sports at different levels of clutch dynamics, one can read Schweickle et al.[18] Some papers like Pelechrinis,[19] Goldner,[20] Palmquist et al.[21] etc. are there that enriches decision making in NFL through rigorous quantitative analysis but the issues covered are in no way related to clutch or pressure situation of the game.

Devising quantitative tools to gauge the performance level of individual players and teams in different ball games is crucial, as the capability of players and teams to navigate high-stress moments has now become an integral part of sports analytics. This has become essential for decision-making in team and individual sports in terms of team combination, substitution decisions, game strategy formation and so on. By revisiting the existing literature, it is ascertained that the issue under consideration has attracted researchers across most sports but not in the case of American football.

Research in American football analytics that are targeted towards the performance of players and teams has traditionally been focused on outcome-based models without explicitly quantifying how pressure builds and fluctuates throughout a drive or game. This gap needs to be covered by developing models that not only account for game outcomes but also provide a dynamic, quantitative understanding of pressure as it unfolds in real-time.

This study addresses this gap by introducing a novel pressure index framework specifically designed for American football. By integrating key game-specific metrics such as time remaining, score differential, field position, and play context, the proposed model offers a refined approach to capturing situational pressure on both offensive and defensive units. Applying this framework to historical NFL data allows for a deeper exploration of how pressure influences game outcomes, contributing to the growing body of literature in sports analytics while offering practical applications for coaching, performance evaluation, and strategic planning.

Methodology

This study proposes a pressure index framework to quantify situational pressure in American football. The primary objective is to introduce a structured methodology for defining pressure indices that account for game-specific variables such as time remaining, score differential, field position, and down progression. Additionally, this framework facilitates a comparative analysis of NFL teams based on their respective pressure index values, providing insights into offensive and defensive strategies under high-pressure conditions. Finally, the study evaluates classification models that leverage the proposed indices to predict drive success, demonstrating the effectiveness of machine learning approaches in capturing the influence of situational pressure on game outcomes. For clarity, we define key American football terms used throughout this study in Table 1.

Table 1. Model performance metrics.

Term	Notation	Description
Time remaining	T_R	The amount of time left in the game.
The score differential	D	The difference in points between the competing teams.
Field position	F	Measured in yards from the opponent's end zone, indicating the distance the offense must cover to score.
Current down	d	The current attempt (out of four) that a team has to advance at least ten yards.
Yards needed for a first down	δ	The remaining distance required to achieve a new set of downs.
Timeouts remaining	T_O	Number of offensive timeouts still available (critical for late-game time management).

We collected play-by-play data from the 2023 NFL season using the `nflfastR` package. Key situational variables include T_R, D, F, d, δ and T_O. To quantify the urgency of scoring, we define two rates. The *Initial Required Scoring Rate* is given by

$$\mathrm{IRSR} = \frac{\mathrm{P_{req,initial}}}{\mathrm{T_{drive}}}, \tag{1}$$

where $P_{\mathrm{req,initial}}$ represents the points needed based on the score differential at the start of each drive ($P_{\mathrm{req}} = |D| + 3$ when trailing, or 3 when leading/ tied, reflecting a one-possession safety margin) T_{drive} is the total drive time. The *Current Required Scoring Rate* is defined as

$$\mathrm{CRSR} = \frac{\mathrm{P_{req,current}}}{\mathrm{T_R}}, \tag{2}$$

where $P_{\mathrm{req,current}}$ follows the same logic for the current game state ($P_{\mathrm{req}}= |D_{current}|+ 3$ when trailing, else 3).

The first offensive pressure index, PIO1 (down-driven pressure), is defined as

$$\mathrm{PIO1} = \left(\frac{\mathrm{CRSR}}{\mathrm{IRSR}}\right) \times \left(1 + \frac{\mathrm{d}}{4}\right). \tag{3}$$

Here, the ratio $\frac{\mathrm{CRSR}}{\mathrm{IRSR}}$ reflects the escalation in scoring urgency, and the factor $1 + \frac{d}{4}$ increases linearly with the current down, accounting for the diminishing number of opportunities (downs).

The second offensive pressure index, PIO2 (field position pressure), is defined as

$$\mathrm{PIO2} = \left(\frac{\mathrm{CRSR}}{\mathrm{IRSR}}\right) \times \exp\left(\frac{\mathrm{F}}{50}\right). \quad (4)$$

In this expression, the exponential term $\exp\left(\frac{F}{50}\right)$ penalizes the offense for starting farther from the opponent's end zone, thereby increasing pressure as F increases.

To synthesize these two dimensions into a single composite measure, we define the *Composite Offensive Pressure Index* (PIO3) as a weighted average of the normalized indices:

$$\mathrm{PIO3} = \mathrm{w}^* \cdot \mathrm{PIO1}_{\mathrm{norm}} + (1 - \mathrm{w}^*) \cdot \mathrm{PIO2}_{\mathrm{norm}}, \quad (5)$$

where $\mathrm{PIO1}_{\mathrm{norm}}$ and $\mathrm{PIO2}_{\mathrm{norm}}$ are scaled to a 0–100 range. The optimal weight w* was determined via grid search optimization (testing values from 0 to 1 in 0.05 increments) to maximize the area under the receiver operating characteristic curve (AUC-ROC) of a logistic regression model predicting drive success. For each candidate weight w, we computed a temporary index $PIO3_{temp} = wPIO1_{norm} + (1 - w)PIO2_{norm}$ and evaluated its predictive power alongside PID and timeouts. The goal of this optimization is to find the alignment between PIO3 and drive success, where "drive success" measures the effectiveness of an offensive series of plays in advancing the ball and scoring points, such as through touchdowns, field goals, or gaining first downs. Here, $PIO3_{temp}$ represents trial combinations during weight optimization, while PIO3 in Equation (5) denotes the final composite index using the optimal weight w^*.

To capture the defensive perspective, we define the *Offense Progress Indicator* (OPI) as

$$\mathrm{OPI} = \frac{\mathrm{YardsGained} + 1}{\mathrm{YardsNeeded} + 1}, \quad (6)$$

where YardsGained is measured relative to the distance required for a first down or touchdown. The addition of 1 to both the numerator and denominator serve as a smoothing factor to prevent division-by-zero errors and mitigate the impact of very small values.

The *Defensive Pressure Index* (PID) is then calculated as

$$\mathrm{PID} = \mathrm{OPI} \times \left[\left(\frac{4 - \mathrm{d}}{4}\right) + \exp\left(-\frac{\mathrm{F}}{100}\right)\right], \quad (7)$$

with the term $\frac{4-d}{4}$ reflecting the number of remaining downs and $\exp\left(-\frac{F}{100}\right)$ increasing defensive pressure as the offense nears the end zone.

The pressure indices were validated using logistic regression, Random Forest, and Gradient Boosting Machine models. In these models, both the composite offensive pressure index *(PIO3)* and the defensive pressure index *(PID)* were used together as predictors, along with their constituent normalized components ($PIO1_{norm}$ and $PIO2_{norm}$) and the number of timeouts remaining. Models were evaluated via an 80/20 stratified train-test split (n = 4,757 training drives, n = 1,189 test drives from 5,946 total 2023 season drives), with Random Forest's internal out-of-bag error used for hyperparameter tuning. This large sample size ensures robust performance estimation without requiring k-fold cross-validation. Timeouts remaining (T_O) were explicitly included as they critically influence late-game pressure dynamics, particularly around the two-minute warning. Variables such as current down and field position were omitted from the models because their effects are already encapsulated within the pressure indices. The target variable was drive success, defined as a binary outcome indicating whether a touchdown was scored during the drive. We focused on touchdowns rather than field goals as they represent more decisive offensive achievements and better reflect sustained drive effectiveness under pressure. Model performance was assessed using metrics such as AUC-ROC, accuracy, precision, recall, and F1-Score, with ensemble methods (Random Forest and GBM) demonstrating superior performance compared to logistic regression.

This integrated framework provides interpretable, dynamic measures of situational pressure by combining offensive and defensive aspects of the game. It facilitates the identification of clutch moments of the match and enhances performance evaluation and strategic decision-making in American football.

Data analysis

Dataset description

The dataset for this study was sourced from the National Football League (NFL) play-by-play data for the 2023 season, retrieved using the nflfastR package in R. The entire analysis pipeline, including data acquisition, preprocessing, and the final analysis, is contained in a single script available on GitHub at https://github.com/agnivibes/nfl-pressure-index. This ensures full reproducibility from data pulling to the final results. NFL is a professional American football league consisting of 32 teams. The league is divided into two conferences viz. AFC (American Football Conference) and NFC (National Football Conference). Each conference has four divisions (North, South, East, and West), with four teams in each. From each conference (AFC and NFC), 7 teams qualify based on their win-loss records. Teams compete in a regular season (18 weeks, 17 games per team), and the best teams qualify for the NFL Playoffs, leading to the Super Bowl, which decides the champion. The relevant play-by-play data from NFL official 2023 standings are collected from the website https://www.espn.com/nfl/standings/_/season/2023. After cleaning and filtering to exclude incomplete or inconsistent records, the dataset focused on situational variables critical to modelling offensive and defensive pressure. These variables included game-

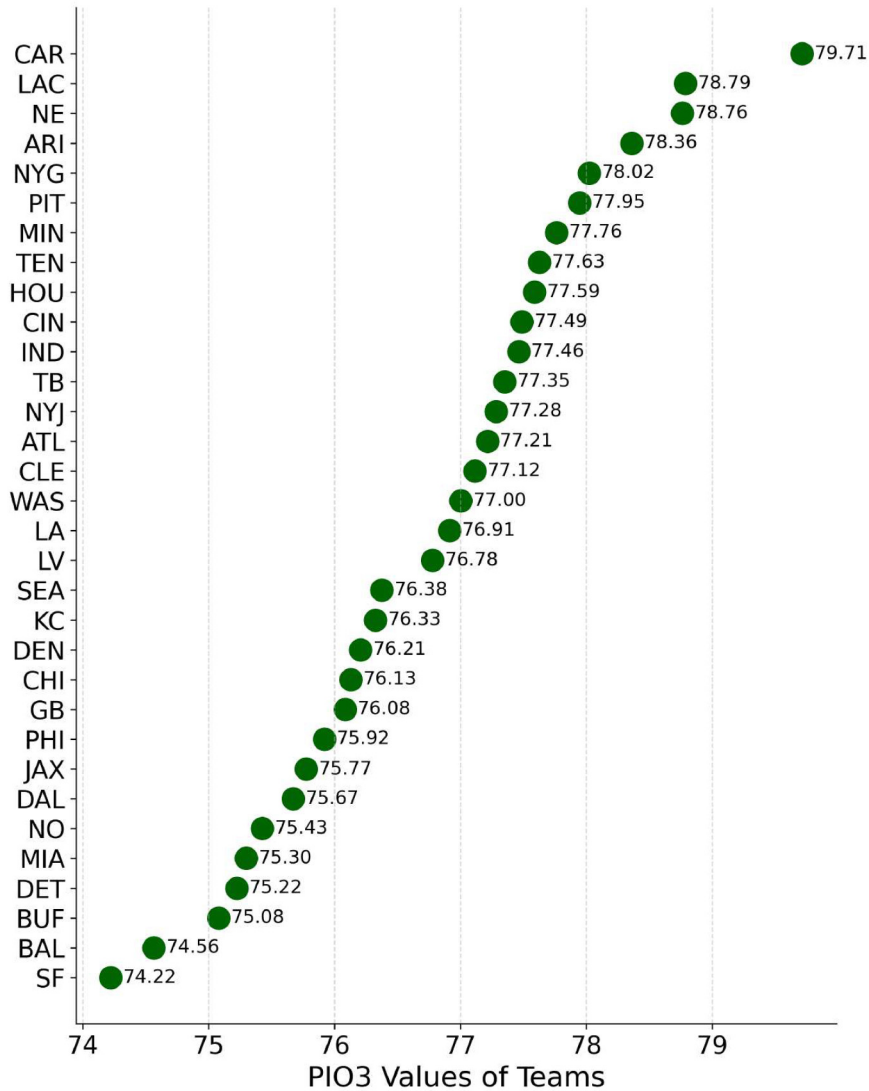

Figure 1. Average Composite Offensive Pressure Index (PIO3) by Team.

seconds remaining, yard line position, down, score differential, and timeouts. Drive-level aggregates were computed to capture dynamic metrics such as drive time, field position, down progression, and scoring requirements, all of which were essential in the construction of the Composite Offensive Pressure Index (PIO3) and Defensive Pressure Index (PID).

Comparative team analysis

To understand how different NFL teams handled offensive pressure during the 2023 season, a team-level analysis was conducted using the Composite Offensive Pressure Index (PIO3) and Defensive Pressure Index (PID).

Offensive pressure (PIO3). Figure 1 displays the average PIO3 for each team, revealing patterns in offensive efficiency under pressure. The Carolina Panthers (CAR) exhibited the highest average PIO3, indicating that they frequently encountered high-pressure offensive situations. This may reflect strategic risks, frequent deficits, or challenges in sustaining drive efficiency. The Los Angeles Chargers (LAC), New England Patriots (NE) and Arizona Cardinals (ARI) also had elevated PIO3 values, suggesting similar offensive pressure trends.

Conversely, the San Francisco 49ers (SF), Baltimore Ravens (BAL), and Buffalo Bills (BUF) demonstrated the lowest average PIO3 values. These teams likely maintained greater control over their offensive drives, benefiting from favorable situational contexts or strategic advantages.

Defensive pressure (PID). Figure 2 illustrates the average PID values across teams. The San Francisco 49ers (SF), Detroit Lions (DET), and Denver Broncos (DEN) recorded

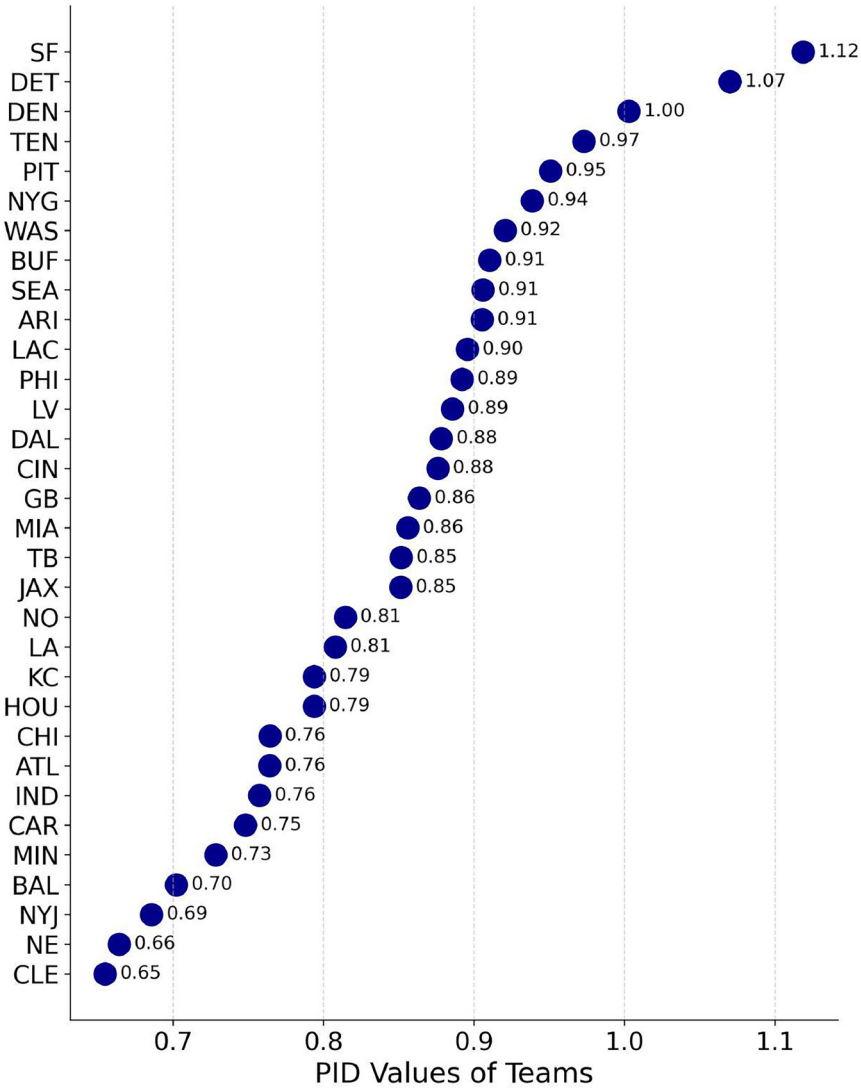

Figure 2. Average Defensive Pressure Index (PID) by Team.

the highest PID scores, indicating strong defensive pressure exerted against opponents. Their ability to impose high defensive pressure may have played a crucial role in disrupting offensive plays and limiting opponent scoring opportunities.

On the other end of the spectrum, the Cleveland Browns (CLE), New England Patriots (NE), and New York Jets (NYJ) displayed the lowest PID values, suggesting comparatively lower defensive pressure. These teams may have faced difficulties consistently disrupting opposing offenses or applying pressure in crucial game situations.

Modeling framework

To evaluate the predictive power and practical utility of the proposed pressure indices, three classification models were employed: Logistic Regression (LR), Random Forest (RF), and Gradient Boosting Machine (GBM). These models incorporated the Composite Offensive Pressure Index (PIO3), Defensive Pressure Index (PID), and team timeouts as primary predictors of drive success, a binary outcome indicating whether a touchdown was scored during the drive. Thereafter, each model was validated using cross-validation and resampling techniques to ensure the robustness of the models.

Model performance evaluation

We evaluated three classifiers namely, Logistic Regression (LR), Random Forest (RF), and Gradient Boosting Machine (GBM),using AUC-ROC (the area under the receiver operating characteristic curve), accuracy, precision, recall, and F1-score (Table 2).

Table 2. Model performance Metrics.

Model	AUC	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.908	88.2%	91.0%	94.2%	92.6%
Random Forest	0.924	89.1%	92.1%	94.0%	93.0%
Gradient Boosting	0.944	90.0%	92.2%	95.1%	93.7%

1. Logistic Regression achieved an AUC of 0.908, accuracy of 88.2%, precision of 91.0%, recall of 94.2%, and an F1-score of 92.6%. Of the three predictors, the Defensive Pressure Index (PID) was the strongest positive driver of drive success ($\hat{\beta} = 1.393$, $p < 0.001$), highlighting how increased defensive pressure boosts the odds of a failed drive for the offense. In contrast, the Composite Offensive Pressure Index (PIO3) was a significant negative predictor ($\hat{\beta} = -1.40$, $p < 0.001$), indicating that greater offensive pressure reduces the likelihood of a successful drive. Finally, each additional timeout remaining also increased drive success ($\hat{\beta} = 0.616$, $p < 0.001$).
2. Random Forest improved on LR, delivering an AUC of 0.924, accuracy 89.1%, precision 92.1%, recall 94.0%, and an F1-score of 93.0%.
3. Gradient Boosting led the pack with an AUC of 0.944, accuracy 90.0%, precision 92.2%, recall 95.1%, and an F1-score of 93.7%, reflecting its ability to capture complex, non-linear interactions.

Figure 3 plots the ROC curves: the GBM curve (red) envelopes RF (green) and LR (blue), underscoring its superior discrimination. Overall, while all models perform strongly, GBM strikes the best balance of sensitivity and specificity, with RF a close second and LR offering the greatest interpretability.

Interpretation and insights

The results highlight the effectiveness of the proposed pressure indices in predicting drive outcomes and understanding team-level dynamics under pressure. Gradient Boosting emerged as the most robust predictive model, striking a balance between sensitivity and specificity. While Random Forest demonstrated comparable performance, logistic regression provided valuable interpretability in understanding the direct impact of pressure variables on drive success.

From a coaching and strategic perspective, teams that frequently encountered high offensive pressure, such as the Panthers and Chargers, may need to refine their play-calling strategies and improve execution in high-pressure moments. In contrast, teams with lower pressure indices

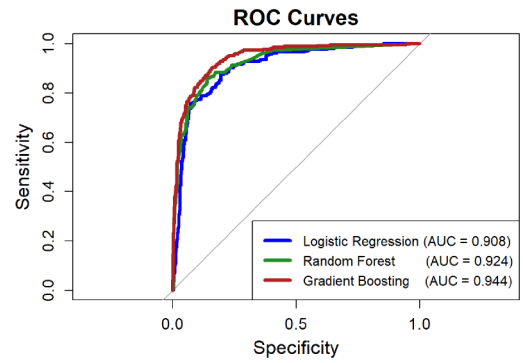

Figure 3. ROC Curves for Logistic Regression, Random Forest, and Gradient Boosting Models.

can optimize their existing strategies to sustain favorable conditions throughout games. Overall, this study introduces a novel, data-driven approach to quantifying situational dynamics in American football. The framework provides actionable insights for performance analysis, coaching strategies, and future sports analytics research.

Conclusion and future work

This study introduces a novel framework for quantifying situational pressure in American foot- ball through dynamic offensive and defensive pressure indices. By integrating critical game- specific variables such as time remaining, score differential, field position, and down progression, we constructed the Composite Offensive Pressure Index (PIO3) and the Defensive Pressure Index (PID). These indices effectively capture the evolving intensity of high-stakes moments, providing interpretable and actionable insights into team performance under varying situational stress.

The team-level analysis of the PIO3 index revealed distinct patterns in how NFL teams handle offensive pressure. Teams such as the Carolina Panthers and Los Angeles Chargers consistently faced high-pressure situations, which could indicate strategic risk-taking, frequent deficits, or inefficiencies in early down execution. In contrast, teams like the San Francisco 49ers and Baltimore Ravens exhibited lower offensive pressure indices, suggesting more effective game management and situational control. Similarly, defensive pressure analysis via PID highlighted teams such as the San Francisco 49ers, Detroit Lions, and Denver Broncos as leaders in using defensive pressure, potentially disrupting opponent offenses more effectively.

A comprehensive model evaluation confirmed that ensemble learning methods significantly outperform traditional logistic regression in predicting drive success. Gradient Boosting achieved the highest predictive accuracy with an AUC of 0.944, closely followed by Random Forest with an AUC of 0.924. The logistic regression model, while offering interpretability, demonstrated comparatively lower

predictive performance (AUC = 0.908). The superior performance of ensemble models suggests that drive success is influenced by complex, non-linear interactions between pressure indices, a factor that tree-based models capture more effectively than logistic regression.

While this framework represents a significant step forward in quantifying pressure in American football, several opportunities for future research remain. Incorporating player-level metrics, such as Next Gen Stats' quarterback 'Time to Throw', receiver separation rates, or defensive coverage schemes derived from player tracking data, could refine the granularity of pressure evaluation by capturing micro-level decision-making under pressure. Additionally, expanding the dataset to include multiple seasons would allow for longitudinal analysis, providing insights into how teams adapt to high-pressure scenarios over time. Future studies could also explore alternative drive-success definitions (e.g., including field goals) to assess how pressure dynamics vary across different scoring outcomes. While our primary focus was quantifying pressure rather than feature selection, future work could employ interpretability techniques (e.g., SHAP values) to analyze how specific pressure indices contribute to predictions under different game contexts. Finally, integrating in-game contextual factors, such as weather conditions, real-time player fatigue metrics (e.g., sprint distance over quarters), or crowd noise levels, may further enhance situational pressure assessment.

The flexibility of this framework allows for adaptation to other sports contexts. In basketball (NBA/WNBA), a pressure index could quantify decision-making during clutch possessions in the final minutes of a game, incorporating shot-clock constraints and defensive coverage. In mixed-martial arts (UFC), the model could assess fighter performance under increasing pressure across rounds, incorporating strike differentials, stamina decay, and grappling effectiveness. In baseball (MLB), situational pressure could be measured in late-inning scenarios, evaluating pitcher-batter dynamics, base-runner impact, and clutch hitting efficiency. These sport-specific adaptations would not only validate the universal applicability of pressure quantification but also provide deeper insight into decision-making and performance across competitive environments.

Overall, this study contributes a systematic and data-driven approach to evaluating situational pressure in American football. By developing interpretable indices and validating their predictive utility through machine learning models, this framework enhances strategic decision-making and performance analysis. Its adaptability across multiple sports further underscores its potential impact in the broader field of sports analytics. Future extensions incorporating richer datasets and additional contextual variables could refine and expand the applicability of this framework, offering a more comprehensive understanding of performance under pressure.

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References

1. Chmait N and Westerbeek H. Artificial intelligence and machine learning in sport research: an introduction for non-data scientists. *Front Sports Act Living* 2021; 3: 682287.
2. Reedy J. *Prime Video's use of AI, Next Gen Stats on NFL games is helping viewers understand the game better*. New York: Associated Press, 2025.
3. Hsu NW, Liu KS and Chang SC. Choking under the pressure of competition: a complete statistical investigation of pressure kicks in the NFL, 2000–2017. *PLoS ONE* 2019; 14: e0214096.
4. Bhattacharjee D and Lemmer HH. Quantifying the pressure on the teams batting or bowling in the second innings of limited-overs cricket matches. *J Sports Anal* 2016; 2: 249–261.
5. Baumeister RF. Choking under pressure: self-consciousness and paradoxical effects of incentives on skilful performance. *J Pers Soc Psychol* 1984; 46: 610.
6. Strauss B. Choking under pressure: positive public expectations and performance in a motor task. *Z Exp Psychol* 1997; 44: 636–655.
7. Sarlis V, Gerakas D and Tjortjis C. A data science and sports analytics approach to decode clutch dynamics in the last minutes of NBA games. *Mach Learn Knowl Extr* 2024; 6: 2074–2095.
8. Otten MP. Choking vs. clutch performance: a study of sport performance under pressure. *J Sport Exerc Psychol* 2009; 31: 583–601.
9. Downey J and McGarrity JP. Pressure and the ability to randomize decision-making: the case of the pickoff play in Major League Baseball. *Atl Econ J* 2019; 47: 261–274.
10. Hibbs D. A conceptual analysis of clutch performances in competitive sports. *J Philos Sport* 2010; 37: 47–59.
11. Pourreza A, Mainer-Pardos E and Nobari H. Implicit learning and football performance under psychological pressure: a narrative review. *Int J Sport Stud Health* 2024; 7: 33–40.
12. Seifried C and Papatheodorou M. The concepts of clutch and choking: recommendations for improving performance under pressure. *J Coach Educ* 2010; 3: 90–98.
13. Arrondel L, Duhautois R and Laslier JF. Decision under psychological pressure: the shooter's anxiety at the penalty kick. *J Econ Psychol* 2019; 70: 22–35.
14. Gu C, Na J, Pei Y, et al. Player pressure map—a novel representation of pressure in soccer for evaluating player performance in different game contexts. *arXiv preprint*. https://doi.org/10.48550/arXiv.2401.16235, 2024.

15. Navia JA, van der Kamp J, Avilés C, et al. Self-control in aiming supports coping with psychological pressure in soccer penalty kicks. *Front Psychol* 2019; 10: 1438.
16. Lemmer HH. A method to measure choking in cricket. *Int J Sports Sci Coach* 2015; 10: 461–477.
17. Shah P and Shah M. Pressure index in cricket. *IOSR J Sports Phys Educ* 2014; 1: 9–11.
18. Schweickle MJ, Swann C, Jackman PC, et al. Clutch performance in sport and exercise: a systematic review. *Int Rev Sport Exerc Psychol* 2021; 14: 102–129.
19. Pelechrinis K. Decision making in American football: evidence from 7 years of NFL data. In: *MLSA@ PKDD/ECML*. https://ceur-ws.org/Vol-1842/paper_09.pdf, 2016.
20. Goldner K. Situational success: evaluating decision-making in football. In: *Handbook of statistical methods and analyses in sports*. https://doi.org/10.1201/9781315166070-8, 2017.
21. Palmquist W, Elmore R and Williams B. Fourth down decision making: challenging the conservative nature of NFL coaches. *DU Undergrad Res J Arch* 2021; 2: 6.

Appendix: Theoretical limits of the pressure indices

Theorem 3.1.1 The Composite Offensive Pressure index (PIO3) attains its bounds as [0, 100].

Proof. The composite offensive pressure index i.e. PIO3 is defined as

$$PIO3 = w \cdot PIO1_{\text{norm}} + (1 - w) \cdot PIO2_{\text{norm}}, \quad \text{with} 0 \leq \text{w} \leq 1 \tag{8}$$

i.e. PIO3 is the weighted average of the normalized values of $\text{PIO1}_{\text{norm}}$ and $\text{PIO2}_{\text{norm}}$, both normalized within the range of 0 to 100 and also the weights add up to 1. Thus, PIO3 shall also range between 0 to 100 being a weighted average of two quantities that range between [0, 100].

Theorem 3.1.2 The Defensive Pressure index PID is a bounded function and it attains its bounds in [0, 1.75].

Proof. From (7) PID is given as

$$PID = OPI \times \left[\left(\frac{4-d}{4}\right) + \exp\left(-\frac{F}{100}\right)\right]$$

The minimum value of OPI is attained when no yards are gained i.e. Yards gained = 0,

$$OPI = \frac{\text{YardsGained(Y)} + 1}{\text{YardsNeeded(N)} + 1} = \frac{0+1}{N+1}$$

now by taking $N \to \infty$, OPImin = 0.

The minimum value of $[(\frac{4-d}{4}) + \exp(-\frac{F}{100})] = T(say)$ is attained when $d = 4$. Since d is the current down so $d = 1, 2, 3, 4$ and as F is the field position, measured in yards from the opponent end zone hence $F \geq 0$.

Thus, for $d = 4$ and $F \to \infty$ the minimum value of T is

$$T_{min} = \left(\frac{4-d}{4}\right) + \exp\left(-\frac{F}{100}\right) = 0 + exp(-\infty)) = 0$$

Thus, $\text{PID}_{\text{min}} = \text{OPI}_{\text{min}} + \text{T}_{\text{min}} = 0 + 0 = 0$.

Again the maximum value of OPI is obtained if the maximum yards that can be gained is achieved i.e. Y = N. Thus,

$$OPI_{max} = \frac{N+1}{N+1} = 1$$

Now T_{max} is attained if $d = 1$ (first down) and $F = 0$ (goal line position) so,

$$T_{\max} = \left[\left(\frac{4-1}{4}\right) + \exp\left(-\frac{0}{100}\right)\right] = 0.75 + \exp(0) = 1.75$$

Thus,

$$PID_{max} = OPI_{max} \times T_{max} = 1 \times 1.75 = 1.75$$

Hence, PID is bounded with bounds [0, 1.75].